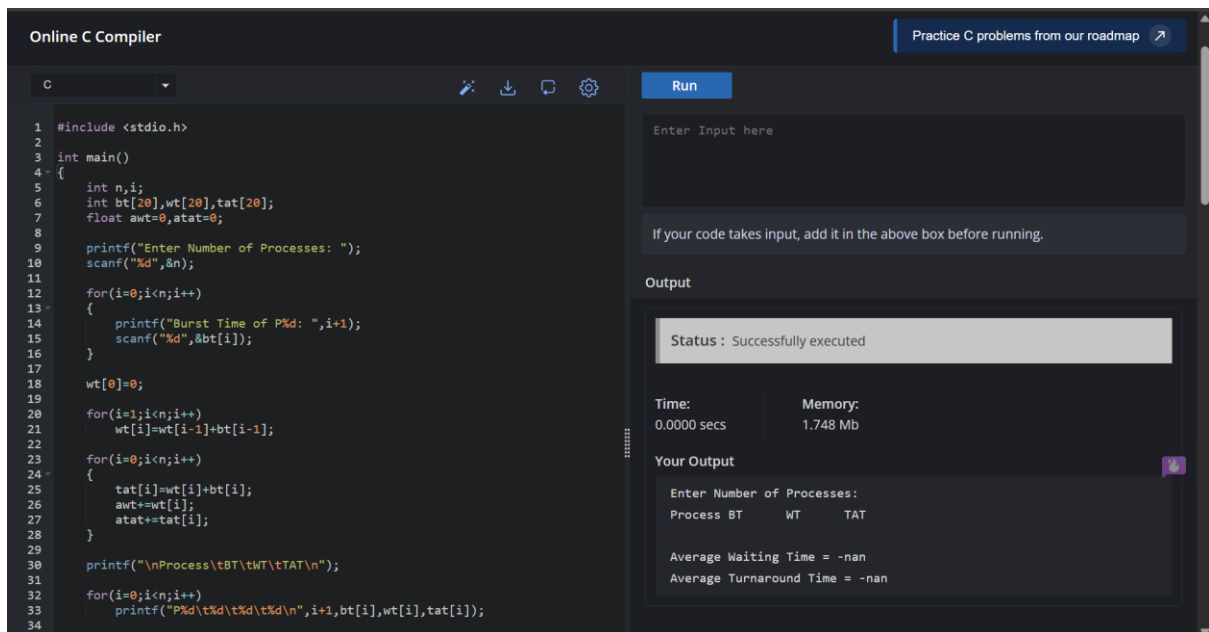


EXPERIMENT 03:

3. Design a CPU scheduling program with C using First Come First Served technique with the following considerations.

- a. All processes are activated at time 0.
- b. Assume that no process waits on I/O devices.



```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n,i;
6     int bt[20],wt[20],tat[20];
7     float awt=0,atat=0;
8
9     printf("Enter Number of Processes: ");
10    scanf("%d",&n);
11
12    for(i=0;i<n;i++)
13    {
14        printf("Burst Time of P%d: ",i+1);
15        scanf("%d",&bt[i]);
16    }
17
18    wt[0]=0;
19
20    for(i=1;i<n;i++)
21    {
22        wt[i]=wt[i-1]+bt[i-1];
23    }
24
25    for(i=0;i<n;i++)
26    {
27        tat[i]=wt[i]+bt[i];
28        awt+=wt[i];
29        atat+=tat[i];
30    }
31
32    printf("\nProcess\tBT\tWT\tTAT\n");
33
34    for(i=0;i<n;i++)
35    {
36        printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
37    }
38
39    printf("\nAverage Waiting Time = %.2f\n",awt/n);
40    printf("\nAverage Turnaround Time = %.2f\n",atat/n);
41}
```

Enter Input here

If your code takes input, add it in the above box before running.

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.748 Mb

Your Output

Enter Number of Processes:

Process	BT	WT	TAT
P1	10	0	10
P2	5	10	15
P3	8	15	23
P4	3	23	26
P5	6	26	32

Average Waiting Time = -nan
Average Turnaround Time = -nan

PSEUDO CODE :

Start

Input number of processes

For each process

 Input Burst Time

Waiting Time of first process = 0

For remaining processes

 Waiting Time = Sum of previous burst times

Turnaround Time = Waiting Time + Burst Time

Calculate Average Waiting Time

Calculate Average Turnaround Time

Display results

Stop

SAMPLE OUTPUT :

Process	BT	WT	TAT
---------	----	----	-----

P1	5	0	5
----	---	---	---

P2	3	5	8
----	---	---	---

P3	2	8	10
----	---	---	----

Average Waiting Time = 4.33

Average Turnaround Time = 7.67CODE:

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n,i;
```

```
    int bt[20],wt[20],tat[20];
```

```
    float awt=0,atat=0;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++)
```

```
    {
```

```
    printf("Burst Time of P%d: ",i+1);

    scanf("%d",&bt[i]);
}

wt[0]=0;

for(i=1;i<n;i++)
    wt[i]=wt[i-1]+bt[i-1];

for(i=0;i<n;i++)
{
    tat[i]=wt[i]+bt[i];

    awt+=wt[i];

    atat+=tat[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for(i=0;i<n;i++)

    printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);

printf("\nAverage Waiting Time = %.2f",awt/n);

printf("\nAverage Turnaround Time = %.2f",atat/n);

return 0;
}
```